

Electrophilic addition reaction

Learning outcomes

At the end of this section you should be able to:

- (R3.4.4) Recognise the electrophile in chemical reactions.
- (R3.4.5) Deduce equations for the reactions of alkenes with water, halogens and hydrogen halides.

The formation of a new covalent bond occurs when nucleophiles attacks electrophiles. This type of reaction is very common in carbon-based reactions such as substitution and addition reactions. How does an addition reaction differ from a substitution reaction? The term 'addition reaction' is explained in [section R3.2.11](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/standard-electrode-potentials-hl-id-46335/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/standard-electrode-potentials-hl-id-46335/>) describing a hydrogen atom that is added to reduce the degree of an unsaturated compound such as alkene or alkynes. How do you identify the electrophile in a chemical reaction?

Electrophiles in chemical reactions

Figure 4 in the [previous section](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/heterolytic-fission-id-46328/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/heterolytic-fission-id-46328/>) shows the hydroxide ion attacking the partially positively charged carbon atom in bromoethane. This type of electron-deficient carbon is called an electrophile. This carbon atom is looking for an electron-rich species to fill its electron-deficient site by forming a new covalent bond. Electrophile means an electron-deficient species (neutral or positively charged atom) and it has a tendency to accept or to attract a pair of electrons from the high electron density region.

What are some common types of electrophile and what types of molecule do they usually react with?

An electrophile has an electrophilic centre atom, often containing an incomplete octet, which carries a partial or full positive charge. These can be protons, carbocations, halogen molecules or hydrogen halides, any molecule that has a carbonyl group and molecules with incomplete octets such as boron trifluoride.

The electrophile as a deficient-electron species seeks an electron pair from the nucleophiles to complete its valence shell. The reaction between a nucleophile and an electrophile occurs in electrophilic addition or substitution reactions. Common examples of electrophiles are shown in **Figure 1**.

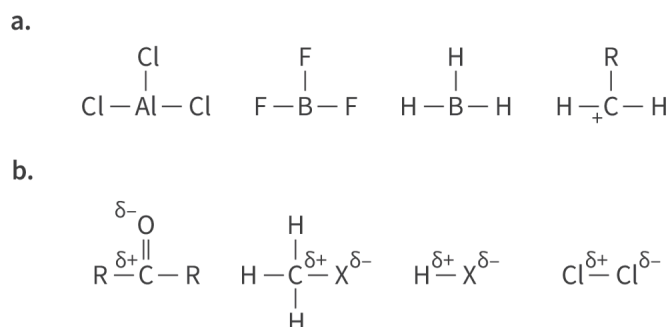


Figure 1. Examples of electrophiles. **a.** Contain atoms with an incomplete octet. **b.** Contain partial positively charge atom.

 More information for figure 1

The diagram shows examples of electrophiles, divided into two parts labeled 'a.' and 'b.'

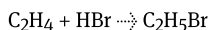
Part 'a.' illustrates chemical structures containing atoms with an incomplete octet: - Cl—Al—Cl, a linear structure with chlorine atoms bonded to aluminum. - F—B—F, a structure with fluorine atoms bonded to boron. - H—B—H, a similar structure with hydrogen atoms bonded to boron. - H—C⁺—H, where carbon is positively charged, bonded to two hydrogen atoms.

Part 'b.' displays structures containing partially positively charged atoms: - R—C(δ^+)—O(δ^-)=C—R, illustrating the carbon and oxygen atoms with delta positive and negative charges, respectively. - H—C(δ^+)—X(δ^-), showing hydrogen, carbon, and a generic atom X with partial charges. - Cl(δ^+)—Cl(δ^-), demonstrating chlorine atoms with partial charges. - H—X(δ^+)—X(δ^-), showing hydrogen bonded to a generic atom X with partial charges.

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An electrophile is a species that can not only accept a lone pair from the nucleophile, but can also initiate the electron movement by attracting the electron-rich species towards itself. The attraction between the electrophile and nucleophile causes the transfer of electrons to occur, leading to the formation of new bonds.

For example, the reaction between ethene and hydrogen bromide (**Figure 2**).



Hydrogen bromide as an electrophile is attracted to the electron-rich double bond of ethene, causing the bond between the two carbon atoms to break. This opens up two new bonding positions, one on each carbon atom. The bond between the H and Br in HBr undergoes heterolytic bond fission. The hydrogen attaches to one of the carbon atoms and the bromide ion attaches to the other carbon atom, producing 1-bromoethane.

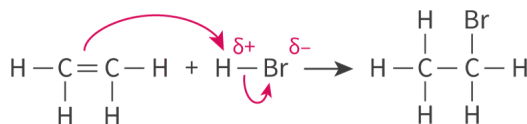


Figure 2. Electrophiles initiate electron movement in addition reactions.

 More information for figure 2

The diagram illustrates the electrophilic addition reaction of hydrogen bromide (HBr) to ethene, forming 1-bromoethane. On the left, the structure of ethene is shown with two carbon atoms double-bonded and each carbon having two hydrogen atoms attached. A curved arrow indicates the movement of electrons from the double bond towards the hydrogen atom in HBr. Additionally, there is a delta positive (δ^+) symbol near the hydrogen and a delta negative (δ^-) symbol near the bromine atom, signifying polarity in HBr.

As the reaction progresses, the bond between the carbon atoms in ethene breaks, and a new bond forms between one of the carbon atoms and the hydrogen (H) from HBr. This results in the other carbon atom forming a bond with the bromide (Br) ion. The final structure on the right shows 1-bromoethane, where one carbon atom is bonded to three hydrogen atoms and the other carbon is bonded to bromine and two hydrogen atoms. The diagram captures the key steps and electron movement involved in electrophilic addition reactions.

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Figure 3 shows an example of how the nucleophile attacks an electrophile. The hydroxide ion (the nucleophile), contains a non-bonding pair of electrons, collides with the carbon atom in carbocation that is positively charged (as an electrophile). The negative electrons are strongly attracted to the positively charged carbon, which then accepts the pair of electrons from the hydroxide ion, forming a coordination bond. Other examples of an electrophile in chemical reactions are shown in **Figure 4**, where the carbon in bromoethane obtains electrons from the hydroxide ion.

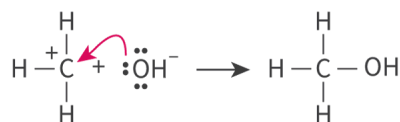


Figure 3. A nucleophile attacks an electrophile.

 More information for figure 3

The image depicts a chemical reaction where a hydroxide ion (OH^-) acts as a nucleophile and attacks a carbocation (C with a positive charge) which is acting as an electrophile. The diagram illustrates the interaction between the negatively charged hydroxide ion, which has a pair of non-bonding electrons, and the positively charged carbon atom in the carbocation. An arrow points from the electrons on the hydroxide ion to the carbon atom, indicating the movement of electrons. As a result, the carbon atom accepts the pair of electrons and forms a new coordination bond with the hydroxide ion, turning into a neutral alcohol molecule, with the hydroxide now bonded to the carbon. The diagram highlights the transformation from a carbocation to an alcohol.

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Figure 4 is another example of a neutral electrophile receiving a pair of electrons from 2-bromo-2-methylpropane and forming a new covalent bond. The bond that is formed between bromide and aluminium atoms is also called a coordination bond where the electrons are shared from one atom. The coordination bond is introduced in [section S2.2.3 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-bonds-id-43594/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-bonds-id-43594/) and is explained in more detail in the [HL section R3.4.7–8 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-complex-hl-id-46330/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-complex-hl-id-46330/).

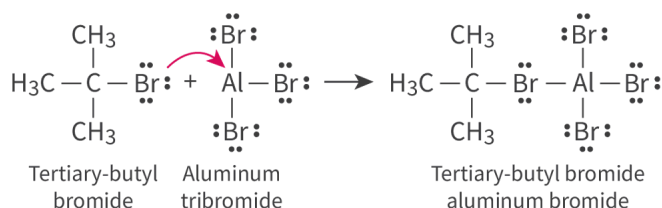


Figure 4. A nucleophile attacks a neutral electrophile.

 More information for figure 4

The diagram shows a chemical reaction illustrating a nucleophilic attack. On the left, there is a structure labeled "Tertiary-butyl bromide" showing a carbon atom connected to three methyl groups (CH_3) and a bromine atom (Br) with lone pairs. This structure is interacting with another compound labeled "Aluminum tribromide," consisting of an aluminum atom connected to three bromine atoms with lone pairs. An arrow indicates the movement of electrons from the bromine atom in the tertiary-butyl bromide towards the aluminum atom in aluminum tribromide. On the right of the arrow, a new structure is formed, labeled "Tertiary-butyl bromide aluminum bromide," showing the tertiary-butyl group now connected to an aluminum-bromine complex through a coordinate bond. This bond is formed as a result of electron sharing from the bromine atom.

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**Exercise 1**

Click a question to answer



1. Which of the following statements is true about an electrophile in electrophilic addition reactions?


☒ Check answers

Reactions of alkenes with water, halogens and hydrogen halides

An electrophile is also involved in addition reactions by attacking the unsaturated bond of unsaturated organic compounds such as alkenes and forming a new compound. The structure of alkenes ensures these molecules have a high electron density due to their double bond because this is susceptible to electrophilic attack. The general addition reaction in an alkene is illustrated in **Figure 5**. Here, 'R' represents either a hydrogen or alkyl group and 'A-B' represents a molecule that is being added across the double bond and forms a single bond.

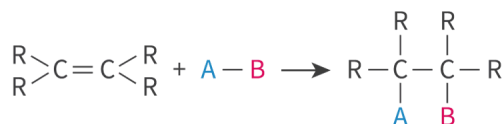


Figure 5. Electrophilic addition reaction of alkenes.

More information for figure 5

The image is a chemical diagram depicting an electrophilic addition reaction of alkenes. It shows a structural formula where a molecule represented as "R"-substituted alkene with a carbon-carbon double bond ($\text{R}-\text{C}=\text{C}-\text{R}$) reacts with a molecule "A-B". After the reaction, the result is a new molecule where "A" and "B" have been added across the former double bond, leading to a single bond structure: $\text{R}-\text{C}(\text{A})-\text{C}(\text{B})-\text{R}$. The text in the diagram uses different colors for "A" and "B" to differentiate them from other elements in the structure, indicating their addition across the double bond.

[Generated by AI]

Alkenes are more reactive compared to alkanes because these unsaturated hydrocarbons contain at least one double bond between the two carbon atoms, and this is where addition reactions such as halogenation, hydrohalogenation and hydration (**Figure 5**) take place. An example is the reaction between ethene and bromine producing 1,2-dibromoethane. The addition reaction in which alkenes are combined with bromine water, $\text{Br}_2(\text{aq})$, is used to test the presence of a double bond in hydrocarbons. This reaction occurs at room temperature quickly.

Making connections

Alkanes contain only a single bond (see [section S3.2.3–4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/homologous-series-id-43865/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/homologous-series-id-43865/>)), making them unreactive towards most chemical reagents such as bromine. Bromine water is orange in colour, and will stay the same when it is mixed with an alkane, but the orange colour fades if its molecules react with the double bonds in alkenes. This colourless solution contains di-halogen alkane that is produced as the double bond is broken. The reaction between alkenes and bromine water is used to differentiate between alkanes and alkenes, and is known as the bromine water test. It is commonly used to identify the presence of double bonds in hydrocarbon samples.

The electrophilic addition of alkenes also occurs with the addition of halogen halide (HX) such as when ethene reacts with hydrogen chloride producing chloroethane (**Figure 5**). The reaction will occur through a concentrated solution of hydrogen halide. Another example of electrophilic addition is hydration, in which alkenes are converted to alcohol by adding water (**Table 1**). Concentrated sulfuric acid is often used to speed up this reaction. This type of reaction is essential as it can be used to produce ethanol on a large scale.

Table 1. Addition reaction examples.

Reactions	Chemical equations
$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}=\text{C}-\text{H} + \text{Br}-\text{Br} \longrightarrow \text{H}-\text{C}-\text{C}-\text{H} \\ \text{Ethene} \quad \text{Bromine} \quad \text{1,2-dibromoethane} \\ \text{Halogenation} \end{array} $ More information	$ \begin{array}{l} \text{C}_2\text{H}_4(\text{g}) + \text{Br}_2(\text{aq}) \rightarrow \text{CH}_2\text{BrCH}_2\text{Br}(\text{aq}) \end{array} $
$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}=\text{C}-\text{H} + \text{H}-\text{Cl} \longrightarrow \text{H}-\text{C}-\text{C}-\text{H} \\ \text{Ethene} \quad \text{Hydrogen chloride} \quad \text{1-chloroethane} \\ \text{Hydrohalogenation} \end{array} $ More information	$ \begin{array}{l} \text{C}_2\text{H}_4(\text{g}) + \text{HCl}(\text{aq}) \rightarrow \text{C}_2\text{H}_5\text{Cl}(\text{aq}) \end{array} $
$ \begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}=\text{C}-\text{H} + \text{H}-\text{OH} \longrightarrow \text{H}-\text{C}-\text{C}-\text{H} \\ \text{Ethene} \quad \text{Water} \quad \text{Ethanol} \\ \text{Hydration} \end{array} $	$ \begin{array}{l} \text{C}_2\text{H}_4(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \end{array} $

Reactions	Chemical equations
👁 More information	$\text{C}_2\text{H}_5\text{OH}$ (l)

The reaction between alkenes and HI is more rapid than with HCl. HI is a stronger acid than HCl, and dissociates more readily producing H^+ and I^- ions in solution (see [section R3.1.6a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strong-and-weak-acids-and-bases-id-44703/>)). Also, HCl has a stronger bond (see [section S2.2.5](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-polarity-id-43853/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-polarity-id-43853/>)) that makes it less ready to react with alkenes. Therefore, the reaction of alkenes and HCl often requires catalysts such as AlCl_3 .

Learning outcome: Demonstrate how to initiate and plan a CAS experience.

How healthy is your diet? A balanced diet is essential not only for your physical but also mental well-being. In recent years, the subject of trans fats  (https://medlineplus.gov/ency/patientinstructions/000786.htm#:~:text=Trans%20fats%20are%20made%20when,from%20vegetable%20oil,in%20a%20reaction%20known%20as%20hydrogenation.) has been hitting the headlines. Trans fats are produced when hydrogen is added to vegetable oil in a reaction known as hydrogenation. In 2015, the Food and Drug Administration  (https://www.fda.gov/food/food-additives-petitions/trans-fat#:~:text=Most%20of%20the%20trans%20fat,This%20process%20is%20called%20hydrogenation.) (FDA) classified trans fats as being no longer generally recognised as safe to eat due to the negative health effects associated with eating them.	
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One possible CAS experience could be to research the guidelines or regulations about the use of trans fats in your country or in a country of your choice. You could also research whether imported foods are subject to the same legislation as locally produced food.

You can check your understanding of addition reactions in alkenes in the activity below.

- **IB learner profile attribute:** Thinker
- **Approaches to learning:** Communication skills — Using terminology, symbols and communication conventions consistently and correctly
- **Time required to complete activity:** 20–30 minutes
- **Activity type:** Pair activity

Your task

There are seven reactions in **Interactive 1** and they are halogenation, hydrohalogenation and hydration of alkene reactions. Arrange the reactants including their reagents and products with a partner by dragging and dropping the structures and reagents into the correct places.

With practise, you will be able to quickly predict the products or reactants of the addition reaction of alkenes.

Reactions	Reactants	Reagent	Products
Halogenation of propene			
Hydrohalogenation of but-1-ene			
Hydration of 2-methylpropene			
Hydrohalogenation of 2-methylpent-2-ene			

$\xrightarrow{\text{H}_2\text{O}}$	$\begin{array}{c} \text{H}_3\text{C} \\ \\ \text{C}=\text{CH}_2 \\ \\ \text{H}_3\text{C} \end{array}$	$\text{H}_2\text{C}=\text{CH}-\text{CH}_2-\text{CH}_3$	$\begin{array}{c} \text{OH} \\ \\ \text{H}_3\text{C}-\text{C}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$
$\begin{array}{c} \text{Br} \\ \\ \text{H}_3\text{C}-\text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$	$\xrightarrow{\text{Br}_2}$	$\begin{array}{c} \text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_3 \\ \\ \text{Cl} \end{array}$	$\xrightarrow{\text{HBr}}$
$\text{H}_2\text{C}=\text{CH}-\text{CH}_3$	$\begin{array}{c} \text{H}_2\text{C}-\text{CH}-\text{CH}_3 \\ \quad \\ \text{Br} \quad \text{Br} \end{array}$	$\xrightarrow{\text{HCl}}$	$\begin{array}{c} \text{H}_3\text{C} \\ \\ \text{C}=\text{CH} \\ \quad \\ \text{H}_3\text{C} \quad \text{CH}_2-\text{CH}_3 \end{array}$

☒ Check

Interactive 1. Arrange the reactants including their reagents and products.

Discuss with your partner the challenges you encounter when arranging these organic compounds. How does this activity help you to understand this section?

Extension task

Use what you learned in [section S2.2.3](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/coordination-bonds-id-43594/>) to name all the products formed from each reaction.